HONEYWELL TECHNOLOGIES CAMPUS CONNECT PROGRAM
HACKATHON 2026

Theme: Cybersecurity | Category: Software

Problem Statement Title
AI-Powered Behavioral Anomaly Detection for Cybersecurity

SOC SENTINEL: Dual-Model Behavioral Anomaly Engine

Technical Architecture & Evaluation Report

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I. Architecture & Methodology:-

SOC Sentinel is an edge-deployable anomaly detection engine built to identify compromised credentials and malicious network behaviors. We deliberately bypassed heavy sequence models (like LSTMs) to ensure low-latency inference on standard entry-level hardware (tested on an Intel Core i3). The system relies on a decoupled architecture: **a Pandas-driven data pipeline**, **a FastAPI Python backend** for asynchronous log streaming, and **a React.js dashboard** for real-time visualization.

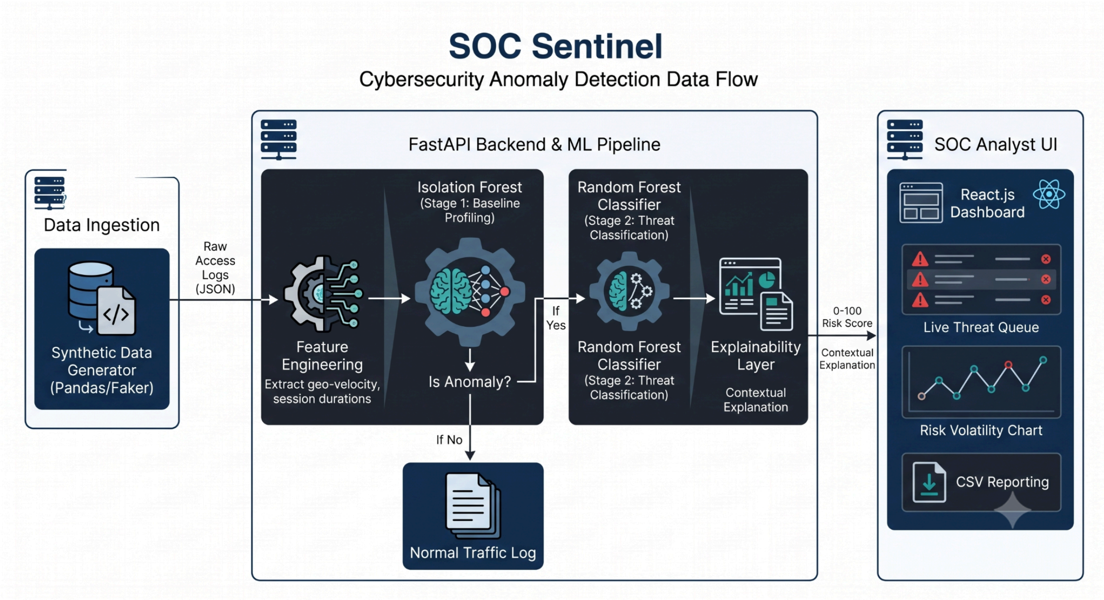

Image 1. Architecture & Pipeline Diagram

II. <u>Synthetic Data Generation & Taxonomy</u>:-

Due to the scarcity of domain-specific, privacy-compliant access logs, a custom Python ***generator*** utilizing Pandas and the Faker library was developed.

- **<u>Schema</u>:** The generator outputs JSON-structured logs containing ***entity_id***, ***entity_type***, ***timestamp***, ***source_ip***, ***resource_accessed***, ***session_duration***, and ***device_fingerprint***.
- **<u>Behavioral Injection:</u>** Normal traffic is simulated using per-entity habitual patterns (consistent IPs and standard business hours). Malicious traffic is injected at a 3-5% frequency, encompassing **Brute Force** (rapid failed authentication), **Impossible Travel** (velocity anomalies), and **Lateral Movement** (accessing unusual/restricted database resources).

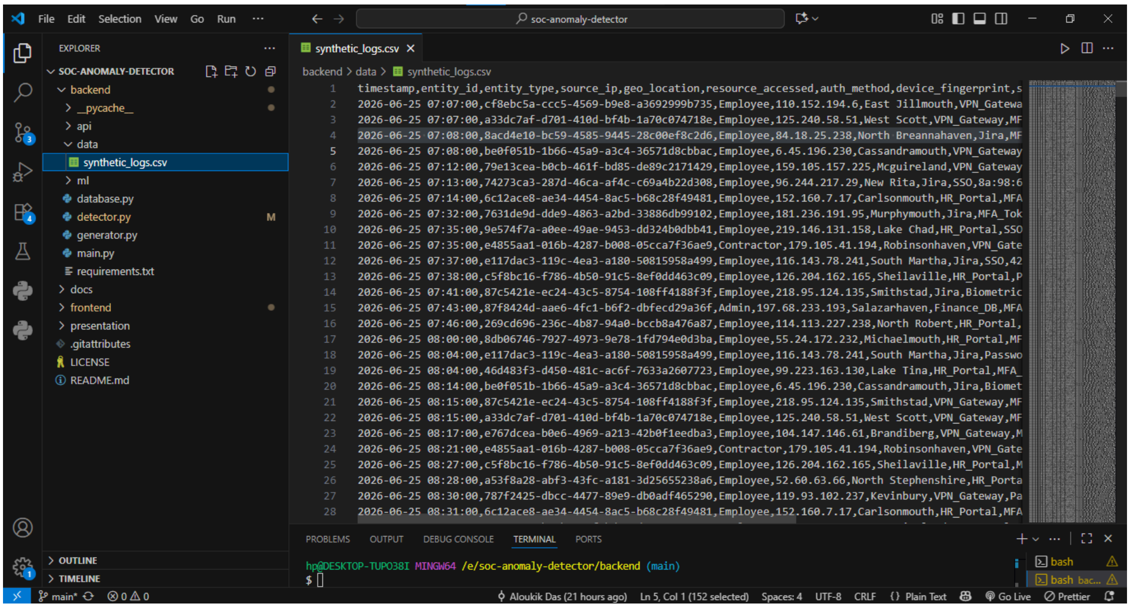

Image 2. synthetic_logs.csv

III. Modeling Strategy:-

The inference pipeline processes incoming log streams sequentially through two distinct models:

- **Baseline Profiling (Isolation Forest):** Acts as an unsupervised filter. It establishes the "normal" behavioral baseline for the enterprise network and calculates a statistical outlier score for every event.
- **Threat Classification (Random Forest):** If the Isolation Forest flags an event (or if the base risk exceeds a designated threshold), the event is passed to a supervised Random Forest classifier. This model categorizes the anomaly into a specific **MITRE ATT&CK** vector.

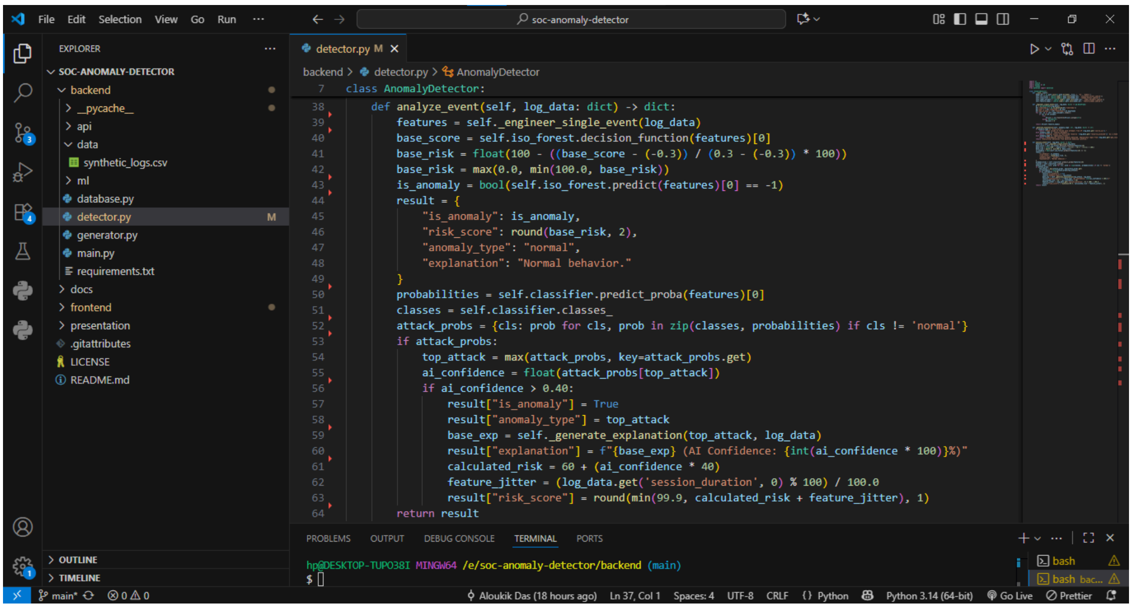

Image 3. detector.py (predict_proba code block)

IV. Explainability & Analyst Usability:-

Security Operations Center (SOC) analysts suffer from alert fatigue when presented with opaque binary alerts. SOC Sentinel addresses this via an Explainability Layer:

- **Dynamic Risk Scoring:** The final 0-100 risk score is not hardcoded. It is a mathematical mapping of the Random Forest's ***predict_proba()*** output.
- **Feature Attribution:** The backend translates the math into human-readable context, appending explanations like *"Geographical velocity anomaly detected. Implausible login from [IP]. AI Confidence: 88%"* to the alert payload.
- **Dashboarding:** The React.js interface features a live queue, dynamic Risk Volatility charting, multi-parameter search/filtering, and one-click CSV export for incident reporting.

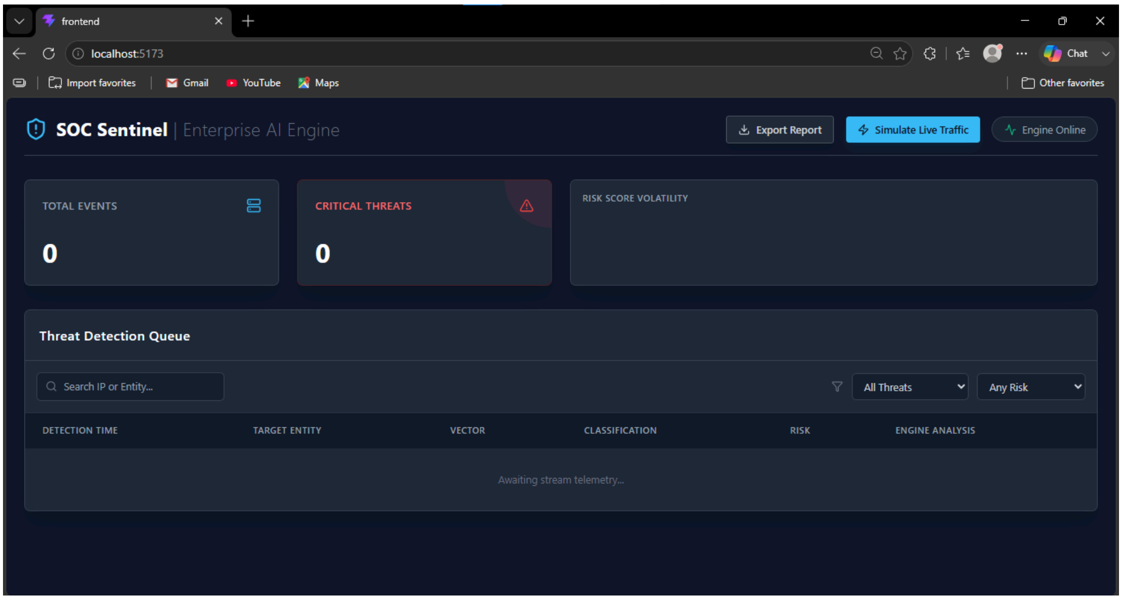

Image 4. Dashboard at startup

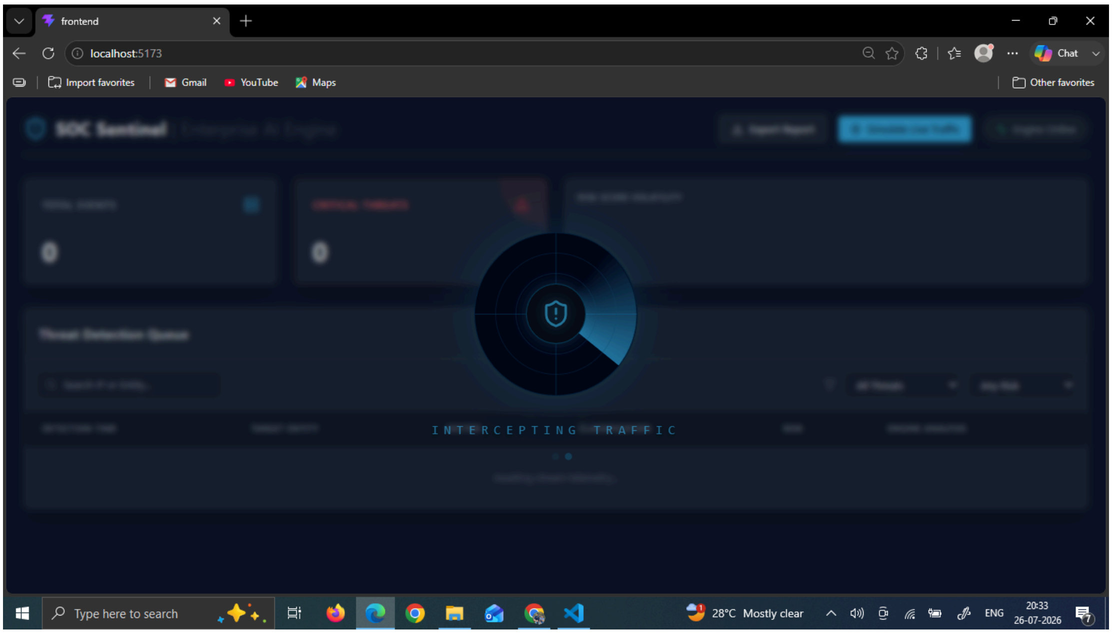

Image 5. Simulation in Process

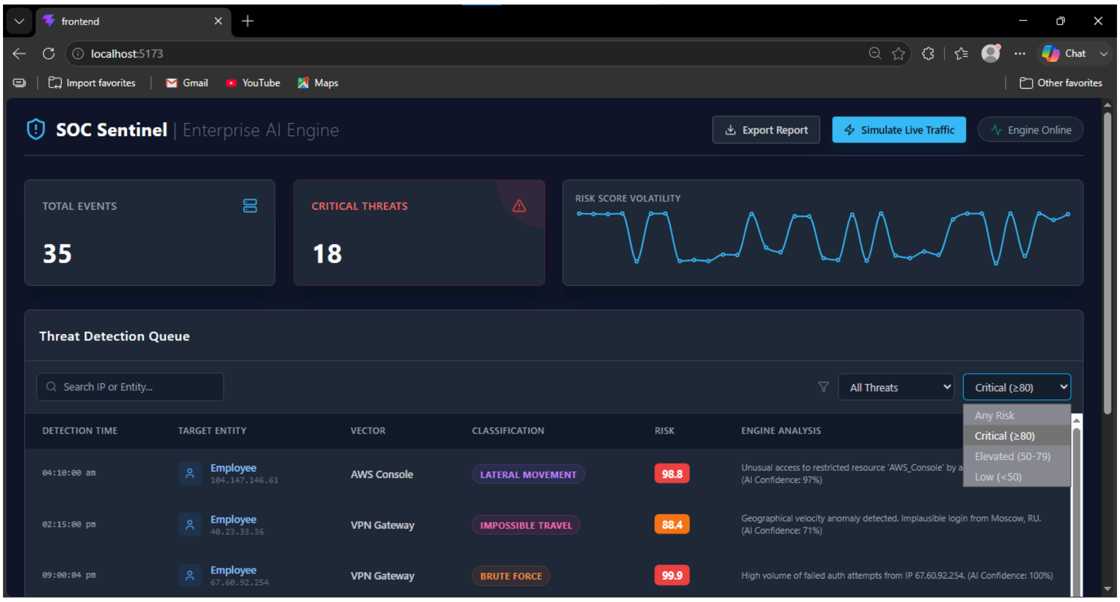

Image 6. Resultant React Dashboard

V. Edge Case Handling:-

- **The Cold-Start Problem:** Traditional sequential models fail when evaluating brand-new employees or devices with no history. SOC Sentinel mitigates this by utilizing the Isolation Forest to score events against the *global* enterprise baseline rather than strictly individual histories. Unseen categorical variables are handled safely via zero-indexed fallback encoders.
- **Concept Drift:** Legitimate enterprise behavior evolves. The lightweight nature of the Scikit-Learn ensemble allows for computationally inexpensive sliding-window retraining. Security teams can retrain the baseline model nightly on the preceding 7 days of logs to adapt to natural behavioral drift.

VI. Model Evaluation & Telemetry Metrics:-

To ensure the system remains viable in a production environment without overwhelming analysts, the ensemble was evaluated using strict thresholding:

- **Precision-Recall Optimization:** Due to the extreme class imbalance (attacks representing <5% of synthetic logs), standard accuracy metrics were discarded in favor of the F1-Score and Area Under the Precision-Recall Curve (AUPRC).
- **False Positive Rate (FPR) Tuning:** The Isolation Forest contamination parameter (***contamination=0.05***) was strictly tuned to limit the baseline alert volume. The system is designed to adhere to a realistic SOC budget, capping alerts at the top 1% of the most statistically anomalous events to prevent analyst alert fatigue.
- **Decision Boundary Confidence:** The Random Forest Classifier evaluates the feature vectors (e.g., Euclidean distance of geo-velocity, access frequencies) and outputs a probability matrix. Only vectors with a mathematically distinct probability threshold (> ***0.40***) override the baseline logic to trigger a critical alert.

--- Classifier Evaluation ---

	precision	recall	f1-score	support
brute_force	1.00	1.00	1.00	41
impossible_travel	0.00	0.00	0.00	2
lateral_movement	1.00	1.00	1.00	5
normal	1.00	1.00	1.00	2365
accuracy			1.00	2413
macro avg	0.75	0.75	0.75	2413
weighted avg	1.00	1.00	1.00	2413

Image 7. Classification Report

VII. Limitations & Future Work:-

Due to hackathon time constraints and local compute limitations, this Minimum Viable Product (MVP) prioritizes pipeline stability over feature bloat.

- Currently, the system models three primary attack vectors. The immediate roadmap includes expanding the taxonomy to detect Device Spoofing and Low-and-Slow Exfiltration.
- The ***command_sequence*** parameter was scoped out of the MVP. Future iterations will integrate a localized transformer model specifically to tokenize and analyze command-line execution sequences for privileged sessions.

VIII. **Executive Summary & Conclusion**:-

SOC Sentinel delivers a highly feasible, edge-ready anomaly detection platform that solves the core challenges of behavioral cybersecurity. By utilizing a dual-model ensemble, we effectively isolated the signal from the noise in highly imbalanced datasets while maintaining the low compute overhead required for industrial gateways and edge devices. Crucially, the system moves beyond black-box AI by providing SOC analysts with dynamic risk scoring and human-readable feature attribution. The result is a proactive, scalable, and explainable defense mechanism capable of intercepting novel threats in near real-time.

IX. **References & Research**:-

1. **Gemini Nano Banana- For image generation of Architecture & Pipeline Diagram of the model.**
2. **Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008).** ***Isolation Forest.*** **IEEE International Conference on Data Mining. (Applied as the theoretical basis for unsupervised baseline outlier detection).**
3. **Scikit-Learn Developers (2024).** ***Ensemble Methods: Random Forests.*** **(Utilized for supervised threat categorization and dynamic risk probability calibration).**
4. **MITRE Corporation.** ***MITRE ATT&CK Framework.*** **(Leveraged as the standard taxonomy for synthesizing Lateral Movement, Brute Force, and Impossible Travel threat vectors).**
5. **NIST Special Publication 800-207.** ***Zero Trust Architecture.*** **(Core principles applied to the continuous behavioral monitoring and scoring of authenticated sessions).**